

SATYAM JAYSAWAL

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SUMMARY

Full Stack Python Developer & AI Engineer with 4 years of experience designing and deploying scalable web applications, Generative AI systems, and machine learning pipelines in production environments. Specialized in end-to-end software development — from RESTful APIs and microservices architecture to Agentic AI, Retrieval-Augmented Generation (RAG) pipelines, and multi-agent LLM orchestration on AWS and Azure cloud platforms. Proficient in Haystack, DSPy, Harness Engineering, Loop Engineering, OpenClaw, Context Engineering, and Prompt Engineering — with hands-on experience building intelligent AI chatbots, agentic workflows, and document processing systems for enterprise-grade SaaS applications. Skilled in MLOps/LLMOps practices including LLM-as-a-Judge evaluation, Verification Loops, Bounded Execution, and AI Observability frameworks. Strong foundation in Data Structures and Algorithms (DSA) with active problem-solving practice on LeetCode, HackerRank, and CodeChef. Proven track record of Agile/Scrum delivery, CI/CD implementation, and cross-functional collaboration across the full Software Development Life Cycle (SDLC).

EDUCATION

- Bachelor of Technology (B.Tech) in Computer Science & Engineering** | Rajkiya Engineering College Mainpuri, AKTU UP | CGPA: 7.57 | August 2018 – July 2022
- Class 12th (Intermediate) – UP Board** | S J G S S I C Sahunagar Gaderiha, Jaunpur, Uttar Pradesh | Percentage: 79.8% | July 2017 – April 2018
- Class 10th (High School) – UP Board** | S M P Y U M V M K B Shankar Ganj, Jaunpur, Uttar Pradesh | Percentage: 83.5% | May 2015 – April 2016

SKILLS

- Programming Languages:** Python, Java, Go (Golang), TypeScript, JavaScript, SQL, HTML, CSS
- Frameworks & Libraries:** Flask, Django, FastAPI, Streamlit, Celery, React JS, Next JS, Node JS, Express JS, Redux, Tailwind CSS
- GenAI & LLM Frameworks:** LangChain, LlamaIndex, Semantic Kernel, Pydantic AI, Hugging Face Transformers, Gradio, Ollama
- LLMs & Foundation Models:** OpenAI (GPT-4o, GPT-4.1, o1/o3 reasoning models), Anthropic Claude (3.5/3.7 Sonnet, Opus), Google Gemini (1.5/2.0 Pro, Flash), Meta Llama 3/4, Mistral
- Agentic AI & Orchestration:** Multi-Agent Systems, ReAct, Tool Use, Function Calling, MCP (Model Context Protocol), FastMCP, Agent2Agent (A2A), Agent Memory, LangGraph Agents, AutoGen, CrewAI, Workflow Orchestration, OpenAI Agents SDK, Google ADK (Agent Development Kit)
- RAG, Vector Database Search & Knowledge Retrieval:** Retrieval-Augmented Generation (RAG), Advanced RAG (Hybrid Search, Re-ranking, GraphRAG, Agentic RAG), Semantic Search, Chunking, Embeddings, FAISS, Qdrant, Pinecone, Chroma, Weaviate, Milvus, pgvector, Azure AI Search
- NLP, Deep Learning & Fine-tuning:** Prompt Engineering, Few-shot/Zero-shot Learning, Fine-tuning (LoRA, QLoRA, PEFT), RLHF, NLP Pipelines
- Multimodal AI & Document Intelligence:** Computer Vision, OCR Pipelines, Azure Document Intelligence, Unstructured.io, Vision-Language Models (VLMs), Image Captioning, Whisper (Speech-to-Text), Audio/Video Transcription, PDF/Document Chat
- AI Evaluation, Observability & Safety:** LLM-as-a-Judge, RAGAS, DeepEval, LangSmith, Promptfoo, LLM Observability (Arize/Phoenix/Arize AI), NeMo Guardrails, Hallucination Detection, PII Redaction, Guardrails AI, Adversarial Testing, Observability & Monitoring
- Cloud AI Platforms (Azure & AWS):** Microsoft 365 Copilot, Copilot Studio, Azure AI Foundry, Azure OpenAI Service (Prompt Flow), Azure Machine Learning, Azure Cognitive Services, Azure AI Vision, Azure Bot Service, Graph API | AWS Bedrock, SageMaker, Kendra, OpenSearch Serverless | GCP Vertex AI, Google AI Studio, Google ADK, GKE, Google Cloud Run
- Cloud, DevOps & Containers:** AWS (EC2, S3, Lambda, ECS), Azure (DevOps/ADO, Pipelines, App Services, Functions, Logic Apps, Blob Storage, SharePoint), Docker, Kubernetes, CI/CD
- Databases:** PostgreSQL, MySQL, MongoDB, CosmosDB, Redis, SQLAlchemy
- Machine Learning & Data Science:** Scikit-learn, TensorFlow, Keras, PyTorch, LightGBM, Pandas, NumPy, Matplotlib, Seaborn, Plotly, OpenCV
- MLOps & LLMOps:** MLflow, Kubeflow, A/B Testing, Guardrails AI
- API Development & Protocols:** REST APIs, GraphQL, gRPC, WebSocket, Microservices, API Integration, Postman
- System Design & Engineering:** Low-Level Design (LLD), High-Level Design (HLD), ETL Pipelines, Event-Driven Architecture, Serverless, Web Scraping
- Version Control & Methodology:** Git, GitHub, GitLab, JIRA, Agile Methodologies (Scrum)
- Java Technologies:** Core Java, Advanced Java, Spring Boot, Spring Batch, Hibernate
- Testing Tools:** Pytest, Selenium, Postman, Regression Testing, Performance Benchmarking
- Data Structures & Algorithms (DSA):** Problem Solving, Competitive Programming | LeetCode, HackerRank, CodeChef

CERTIFICATION

- [AWS Cloud Practitioner](#)
- [Azure Administration Associate\(AZ 104\)](#)
- [Agile Software Development](#)
- [JSP Servlet JDBC Udemy Certificate](#)
- [Spring-Batch Processing with Spring-Boot](#)
- [Python PCEP Certificate](#)
- [Python, Flask, andSQLAlchemy](#)
- [Google's Gemini AI with Streamlit](#)
- [Scalable Web Applications with Python Flask and SQLAlchemy](#)
- [Learn Google's Gemini and Anthropic's Claude API with Python](#)
- [Azure AI Studio \(AI Foundry\) : Prompt Flow, LLMOps & RAG](#)
- [Microsoft Azure AI \(AI-900\)](#)
- Certificate course on Enhancing Soft Skills And Personality from NPTEL

WORK EXPERIENCE

Deloitte USI (AUG 2025 – PRESENT)

Project: Career-Orchestrator Multi-Agent Project

Role: AI Engineer & Full Stack Developer

- Architected a production-grade hierarchical multi-agent career intelligence platform using Google ADK with Meta Supervisor + Sub-Supervisor architecture, Stateful Agent Memory, self-critique loops, and HITL capabilities; built core agents (Supervisor/Orchestrator, O*NET Occupation Deep Profiler, Skills & Knowledge Matcher, Interest Profiler (RIASEC), BLS Economic & Compensation, Career Pathway Planner, Personal Career Match, Location & COL, Output Synthesizer & QA) integrating O*NET, BLS OEWS, USAJOBS.gov, CareerOneStop, and Tavily for Career Transition Planning, Relocation Decision Support, Resume ATS Optimization, and Automation Risk Assessment.
- Implemented Advanced RAG pipelines using LangChain with semantic and hybrid search, Guardrails, PII redaction, A2A, MCP, ServiceNow, and Now Assist Bot for enterprise workflow automation; built hybrid search and re-ranking pipeline combining keyword and semantic search with ML-based match scoring across resumes and jobs.
- Led end-to-end development of TalentVerse, a public-sector intelligent hiring platform with React.js/Redux frontend and Python (FastAPI) microservices, containerized with Docker, CI/CD, and Redis caching, automating recruitment from job creation and candidate matching to career development.

- Engineered intelligent document processing using Claude (via Vertex AI) with OCR to auto-parse resumes and job files into structured candidate/employer profiles; implemented GenAI-driven salary benchmarking, location strategy, benefit recommendations, AI-based skill gap analysis, and personalized learning path suggestions.
 - Built real-time salary calculator with location/experience-based estimates and AI career pathing engine analyzing candidate skills and application history to recommend next-step roles and upskilling paths; developed multi-stage candidate pipeline with AI-driven screening to auto-evaluate applications, generate disposition rationale, and flag top-fit candidates with explainable match scores.
 - Designed AI-powered interview orchestration with LLM-generated role-specific questions, automated panel scheduling, and GenAI scorecard summarization; developed recruiter copilot leveraging RAG and LangChain agents to draft job descriptions, generate Boolean search strings, summarize candidate profiles, and deliver data-driven hiring insights from funnel analytics.
 - Designed REST API and WebSocket-based real-time communication layers with OAuth/JWT authentication; built LSTM and STM sequential prediction models with Docker containerization and CI/CD on GKE; integrated WebSocket for real-time notifications and automated email workflows for candidate engagement.
 - Established observability, monitoring, logging, and tracing with LLM Judge, Eval, and Ground Truth-based evaluation frameworks; used PostgreSQL and BigQuery for structured data storage, analytics, data governance, and automated data quality monitoring.
- Tech Stack: Python, FastAPI, Reactjs, TypeScript, Redux, LangGraph, LangChain, Google ADK, A2A, MCP, Stateful Agent Memory, Semantic Search, Hybrid Search, REST API, WebSocket, OAuth/JWT, Guardrails, PII Redaction, ServiceNow, Now Assist Bot, GCP (BigQuery, GKE, Vertex AI), PostgreSQL, Redis, Docker, Pandas, LSTM, STM, Observability, Monitoring, Logging, Tracing, LLM Judge, Eval, Ground Truth, HITL, O*NET API, BLS Data, USAJOBS API, Tavily, Claude Code CLI, Anthropic (Claude Playground).

Capgemini-India (DEC 2022 - AUG 2025)
Project: H&K Holland & Knight - AI Chatbot Development
Role: AI Engineer & Full Stack Developer

- Developed an AI-powered multi-agent chatbot for Microsoft Teams with LangGraph-based orchestration and Azure OpenAI (GPT-4o) with intent classification and Prompt Flow to dynamically route queries to domain-specific agents (DB, Web Search, Legal KB, Jira MCP integration) for ITSM knowledge retrieval; built scalable backend with FastAPI, Azure Functions, and CosmosDB using LangChain for prompt orchestration and LangGraph for graph-based agentic workflows with tool-calling, conditional routing, and agent memory.
 - Built data pipelines to extract and index SharePoint documents via Azure AI Search (Cognitive Search) for hybrid keyword and semantic retrieval with re-ranking; leveraged Microsoft Graph API to access Teams messages, calendar events, and SharePoint content for real-time context injection; connected to KB via Now Assist Bot Interconnect, delivering structured responses through adaptive cards with contextual follow-ups.
 - Implemented Azure Entra ID (AAD) OAuth 2.0 for secure user authentication and RBAC; automated event-driven workflows via Azure Logic Apps, Azure Service Bus, and Azure Document Intelligence for OCR-based ingestion and structured data extraction.
 - Established CI/CD pipelines using Azure DevOps (ADO) with automated build, test, and deployment stages; containerized services with Docker, Azure Container Registry (ACR), and deployed via Azure Container Apps.
 - Integrated Azure Monitor, Application Insights, and Power BI dashboards for end-to-end observability — tracking agent performance, query latency, query routing accuracy, error rates, and user engagement; used Azure Content Safety for prompt injection mitigation and harmful content filtering.
 - Extended platform via Microsoft Copilot Studio for no-code bot extensions and Microsoft 365 Copilot integration for enterprise-wide AI assistant deployment.
- Tech Stack: FastAPI, Azure Functions, CosmosDB, App Service, Azure Container Apps, Azure Container Registry (ACR), LangGraph, Azure Prompt Flow, MCP (Jira), Unstructured.io, Bot Service, Azure AI Search, Power BI, Logic Apps, Service Bus, Blob Storage, ServiceNow, Document Intelligence, Azure OpenAI (GPT-4o), Azure Entra ID (AAD), Microsoft Graph API, Copilot Studio, Microsoft 365 Copilot, SharePoint, Azure DevOps (ADO), Azure Monitor, Application Insights, AI Guardrails, Prompt Injection Mitigation, PII Redaction, Content Safety

Project – Skoda AI-Powered Chatbot & Document Processing Suite
Role: User Support & AI Developer

- Built AI-powered chatbot, document analyzer, and verifier using Flask (Jinja2) with AWS Bedrock (Claude) and OCR pipelines to auto-parse PDFs, scanned images, and multi-page documents — extracting structured entities, key information, and semantic summaries into structured profiles; deployed event-driven processing via AWS Lambda for serverless document ingestion and async task execution.
 - Implemented Advanced RAG pipeline with FAISS (hybrid search, embeddings, chunking) for "Ask to Files" and BeautifulSoup for "Ask to URL"; built hybrid search and re-ranking with Amazon Kendra and OpenSearch Serverless using ML-based match scoring and LangChain for multi-turn GenAI responses.
 - Built GenAI-driven document insights and automated summarization using AWS Bedrock (contextual recommendations, entity comparisons, structured reports); engineered multi-stage document screening pipeline with auto-evaluation, disposition rationale, and explainable match scores; leveraged SageMaker for fine-tuned NLP model hosting and AWS Lambda for scalable serverless inference triggers.
 - Developed Vehicle Damage Detection POC using YOLOv8 and OpenCV with multimodal capabilities — image captioning, entity extraction, and document comparison.
- Tech Stack: Python, Flask, HTML (Jinja2), AWS (Bedrock, Lambda, Kendra, OpenSearch Serverless, SageMaker, S3), Claude (Anthropic), LangChain, FAISS, OCR, Document Intelligence, YOLOv8, OpenCV, BeautifulSoup, Hybrid Search, Semantic Search, Embeddings.

Project: Amplifier Knowledge Mining & GenAIHUB
Role: Python Developer & AI Engineer

- Architected a knowledge mining and intelligent data retrieval platform with end-to-end ETL/ELT data pipelines using PySpark, Spark SQL, Pandas, and SQL for data ingestion, cleansing, transformation, and loading into a centralized Data Lakehouse on Databricks; performed comprehensive EDA, feature engineering, and statistical analysis on massive datasets.
- Orchestrated complex data and ML workflows through DAGs; built scalable data engineering pipelines with automated schema validation, data lineage tracking, and partitioning strategies; containerized services with Docker and automated deployments via Jenkins CI/CD.
- Established strong data governance frameworks, automated data quality monitoring, anomaly detection, and feedback-driven optimization loops; implemented metadata management and audit logging for end-to-end data traceability.

PERSONAL PROJECTS

- E-commerce Platform** — Built a full-stack multi-role platform (FastAPI, React.js, Next.js, PostgreSQL, SQLAlchemy, NumPy, Pandas) with JWT auth, payment gateway, chatbot UI, analytics dashboard, and real-time notifications for Admin/Vendor/Customer workflows. | [GitHub : https://ecommerce-website-reactjs-vite-frontend.onrender.com](https://ecommerce-website-reactjs-vite-frontend.onrender.com)
- MakeMyTrip App** — Developed a full-stack travel booking app (React.js, Redux, Node.js, Express.js, MongoDB, Tailwind CSS) with JWT auth, Razorpay payments, Nodemailer notifications, session handling, and real-time flight/hotel booking; deployed on Vercel. | <https://make-my-trip-frontend-shi.vercel.app>
- CRUD Operations (Multi-Tech)** — Built CRUD systems across Angular JS, Next.js, Servlet, JSP, Struts, Spring Boot, Spring Security, Spring Batch, Hibernate, JDBC, and MySQL with Bootstrap UI. [GitHub](#)
- MovieFlix** — Developed a movie discovery platform (Flask, MongoDB, Pandas) with a personalized recommendation engine, secure auth, and dynamic search; deployed on Vercel. | <https://movie-flix-app-dataset.vercel.app>
- Personal Portfolio** — Built a responsive portfolio (HTML, CSS, JavaScript) with day/night mode, dynamic color themes, and cross-device media queries. [GitHub : https://satyam-portfolio-q196.onrender.com/](https://satyam-portfolio-q196.onrender.com/)

Applied AI, ML & GenAI Projects

- Handwritten Digit Recognition-**
Developed a Handwritten Digit Recognition system using Convolutional Neural Networks on the MNIST dataset, achieving 98%+ accuracy with TensorFlow and Keras.
- Movie Recommendation System-**

- Built a Movie Recommendation System leveraging collaborative and content-based filtering techniques on the MovieLens dataset using Python, Pandas, and Scikit-learn.
- **Spam Email Classifier-**
Designed a Spam Email Classifier using Naive Bayes and Logistic Regression with TF-IDF vectorization, achieving high precision on the SMS Spam Collection dataset.
- **House Price Prediction-**
Implemented a House Price Prediction model using XGBoost and Random Forest regression with advanced feature engineering on the Ames Housing dataset.
- **Financial News Sentiment Analyzer-**
Built a Financial News Sentiment Analyzer on the Kaggle Financial PhraseBank dataset using FinBERT and LSTM to classify market sentiment from news headlines for trading signals.
- **Face Recognition Attendance System-**
Engineered a Face Recognition-based Attendance System using OpenCV and FaceNet, deployed through a Flask web interface for real-time attendance logging.
- **Real-Time Object Detection-**
Implemented a Real-Time Object Detection system using YOLOv8 on webcam and edge devices, optimized with PyTorch and ONNX for low-latency inference.
- **Stock Price Forecasting-**
Developed a Stock Price Forecasting model combining LSTM and Transformer-based attention mechanisms, visualized through an interactive Plotly dashboard.
- **AI-Powered Image Generation App-**
Built an AI-Powered Image Generation App using Stable Diffusion and the Diffusers library, served through a Gradio UI with a FastAPI backend.
- **Fraud Detection System-**
Developed a Fraud Detection System for imbalanced credit card data using XGBoost with SMOTE oversampling and SHAP-based explainability dashboards.
- **Multimodal Gemini AI App-**
Built a Python and Streamlit app with AI features: chatbot, image captioning, text embedding, PDF chat, and voice assistant. Added audio and video transcription using Google Generative AI and FAISS. <https://github.com/satyamjaysawal/Multiple-Model-RAG-Usecases.git>